AD7190

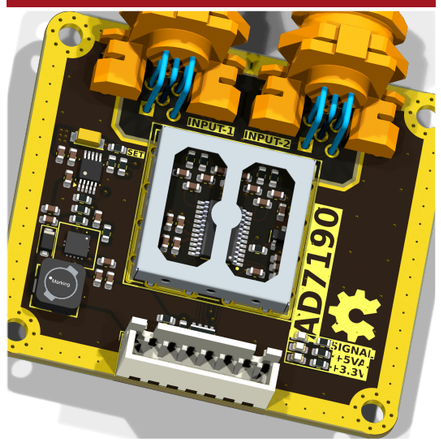
Variant:

2026-10-24

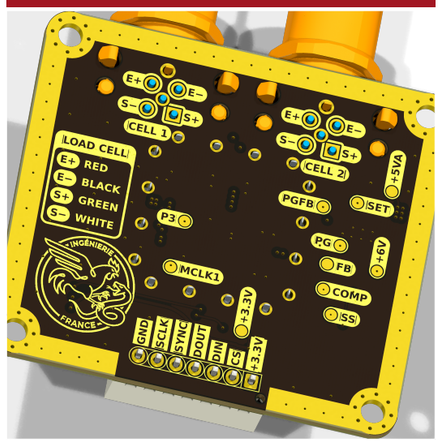
Rev + (Unreleased)

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11	.....	21	.....	31	.....
2	Block Diagram	12	.....	22	.....	32	.....
3	Project Architecture	13	.....	23	.....	33	.....
4	AD7190_Weight_Sensor	14	.....	24	.....	34	.....
5	LT3042_6V_to_5VA	15	.....	25	.....	35	.....
6	TPS61086_3.3V_to_6V	16	.....	26	.....	36	.....
7	Holes, Fiducials	17	.....	27	.....	37	.....
8	Connectors	18	.....	28	.....	38	.....
9	Power - Sequencing	19	.....	29	.....	39	.....
10	Revision History	20	.....	30	.....	40	.....

TOP VIEW



BOTTOM VIEW



NOTES

OPEN_WEIGHT - AD7190 24-bit load-cell front-end, part of OPEN THRUST LAB.
Single +3.3 V host supply, SPI interface, 2 x 4-wire bridge inputs.

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 24-Oct-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
+3.3V host -> TPS61086
6 V -> LT3042 +5VA (AVDD + excitation)

DEBUG NOTE:
TP on every rail.
Wait >= 600 ms (PG) before ADC init.

DESIGN NOTE:
SPI mode 3,
SCLK <= 5 MHz,
RDY on DOUT.

DESIGN NOTE: REFIN1 = bridge excitation: ratiometric.

LAYOUT NOTE: AFE under the WE-SHC shield H5, boost SW node away.

Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: ea7f156
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
Sheet Title: OPEN_WEIGHT - AD7190	File Name: OPEN_WEIGHT.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 9